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(54) **CLOSED COMPARTMENT WELL TOOL FOR WELLHEAD RE-PACKING**

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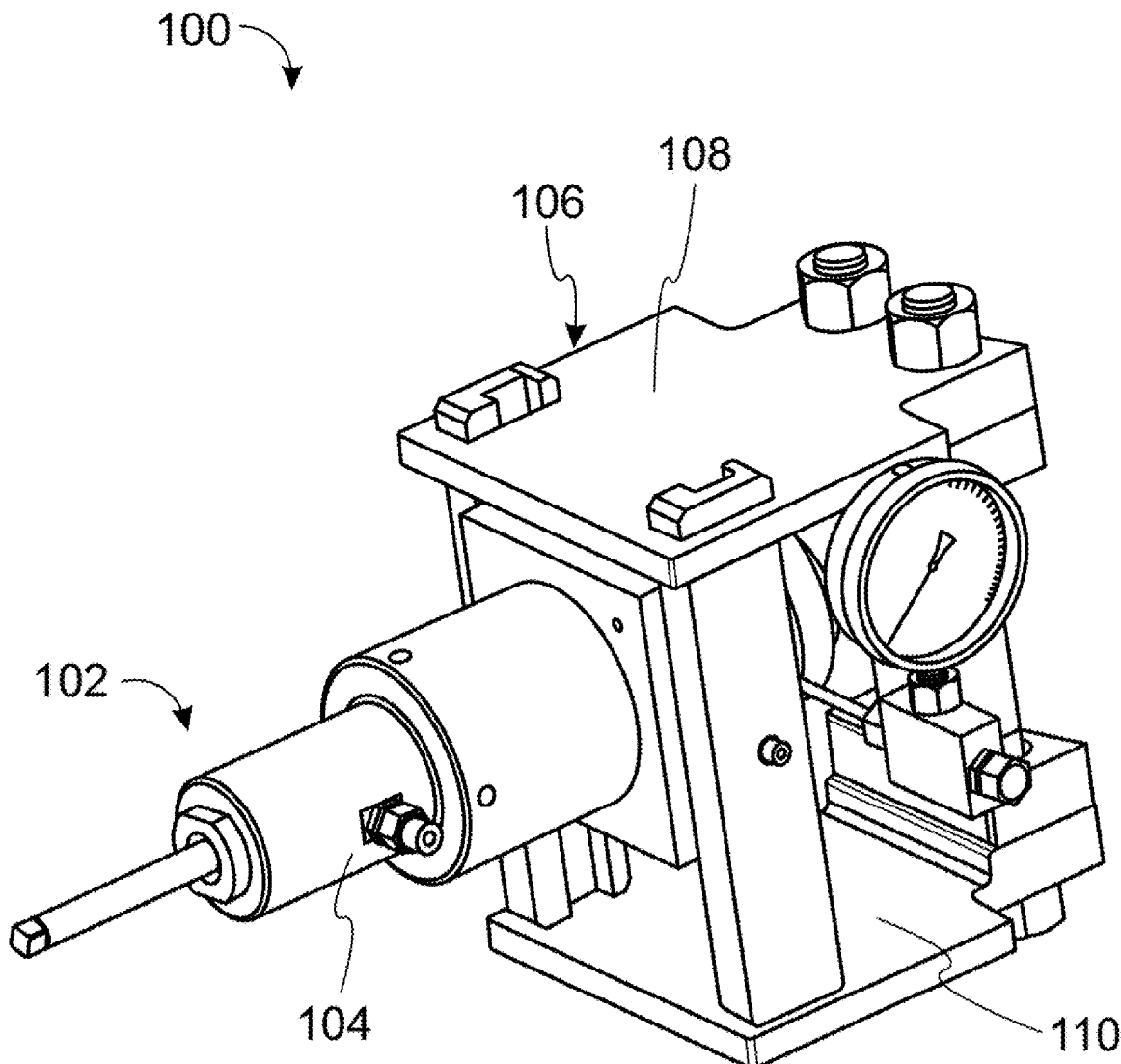
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(60) Provisional application No. 63/558,306, filed on Feb. 27, 2024.

(57) **ABSTRACT**
A well tool assembly well tool includes a coupling subassembly and a supporting subassembly coupled to the coupling subassembly. The coupling subassembly is configured to couple to a pressure port positioned on an above-surface portion of a wellhead of a wellbore. The coupling subassembly includes a body configured to sealingly couple to the pressure port on the wellhead. The supporting subassembly includes a first plate and a second plate, each configured to couple to the portion of the wellhead above and below the pressure port, respectively. The supporting subassembly is configured to orient the body between the first plate and the second plate to couple to the pressure port and to support a weight of the coupling subassembly.



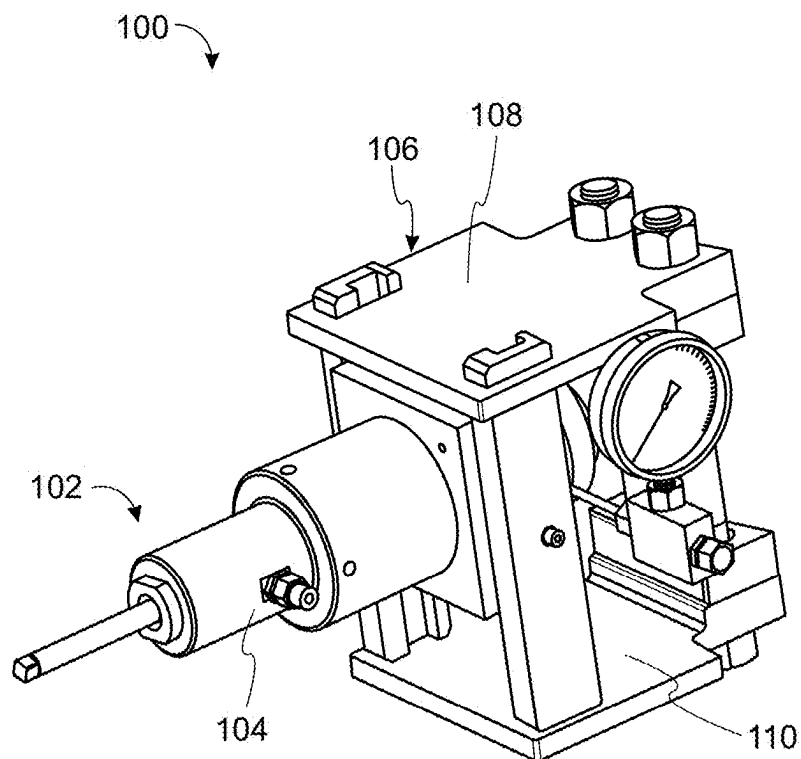


FIG. 1

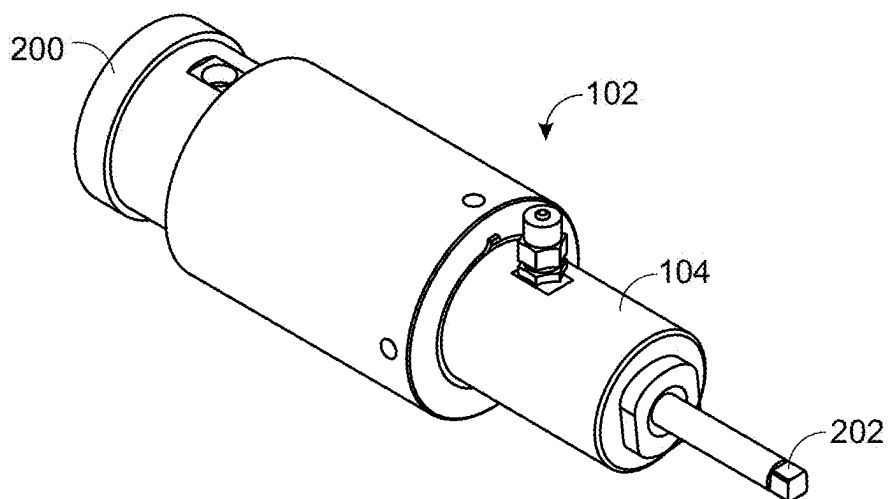


FIG. 2

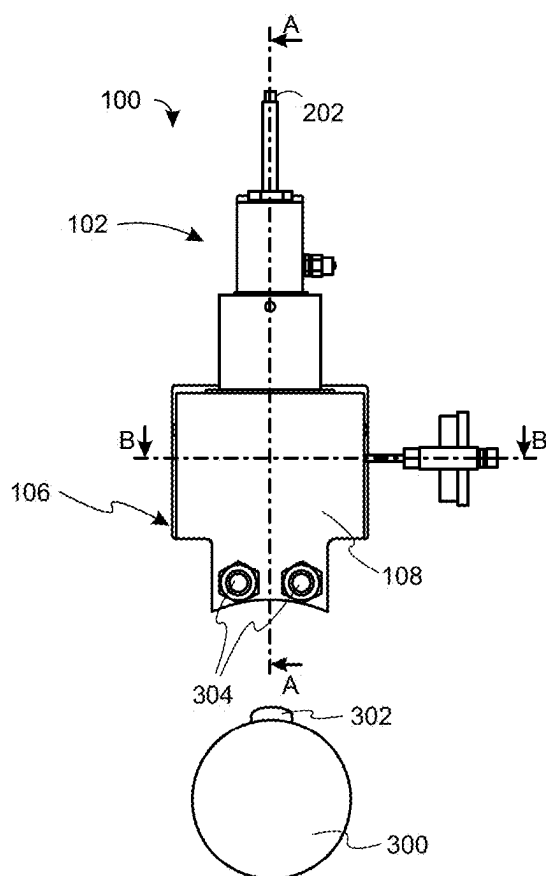
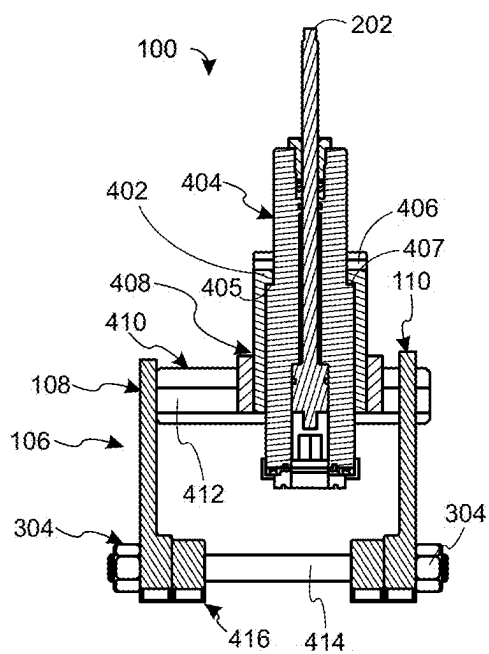
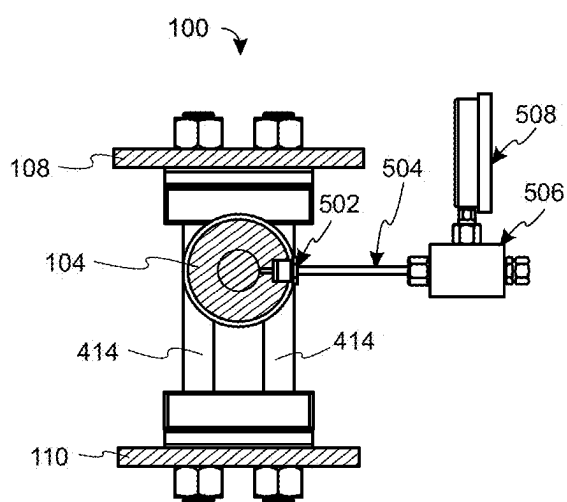


FIG. 3



SECTION A-A

FIG. 4



SECTION B-B

FIG. 5

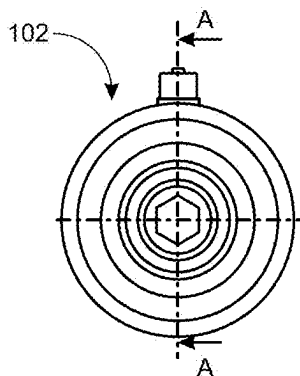


FIG. 6

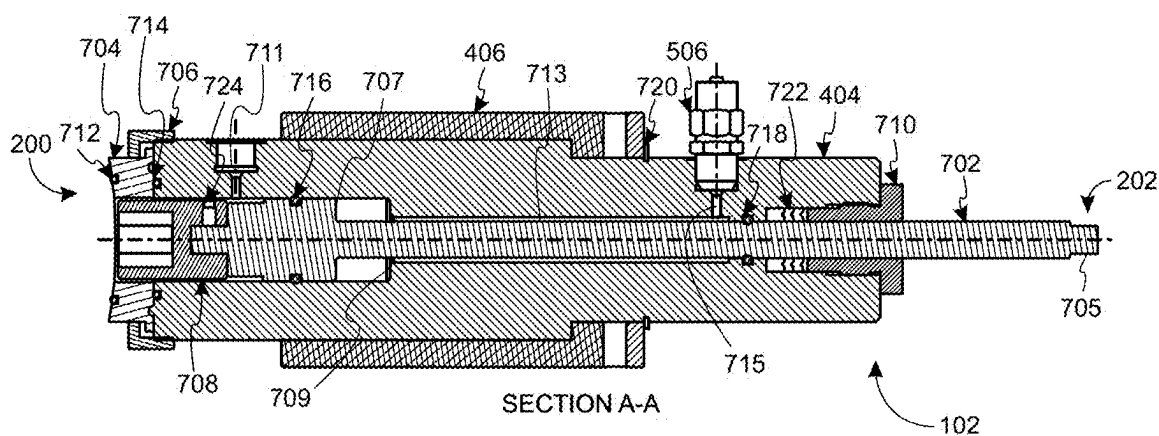


FIG. 7

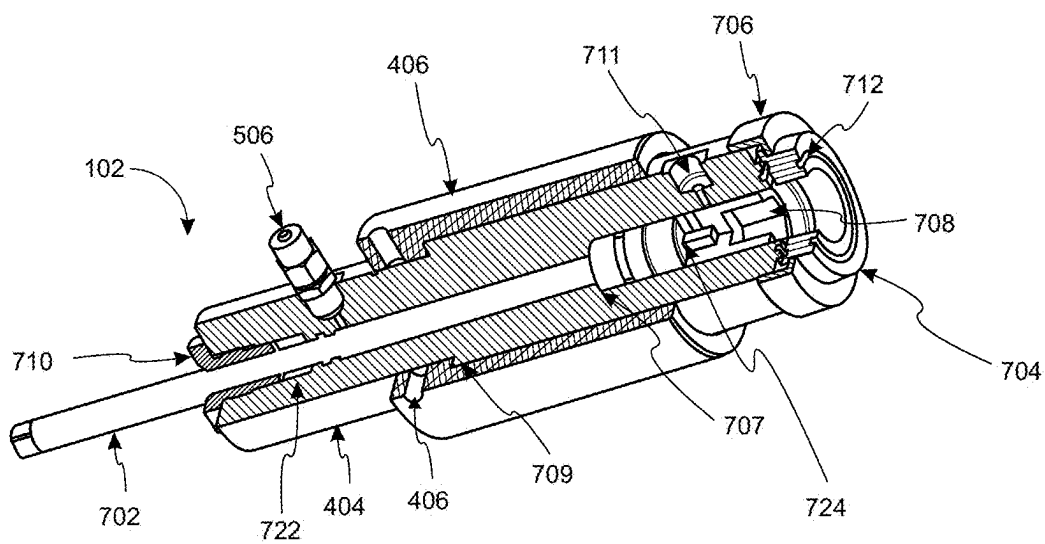


FIG. 8

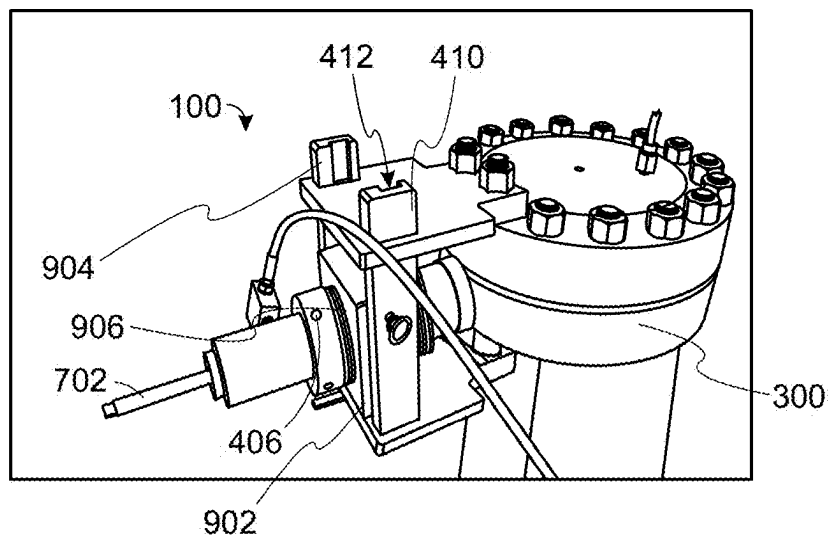


FIG. 9

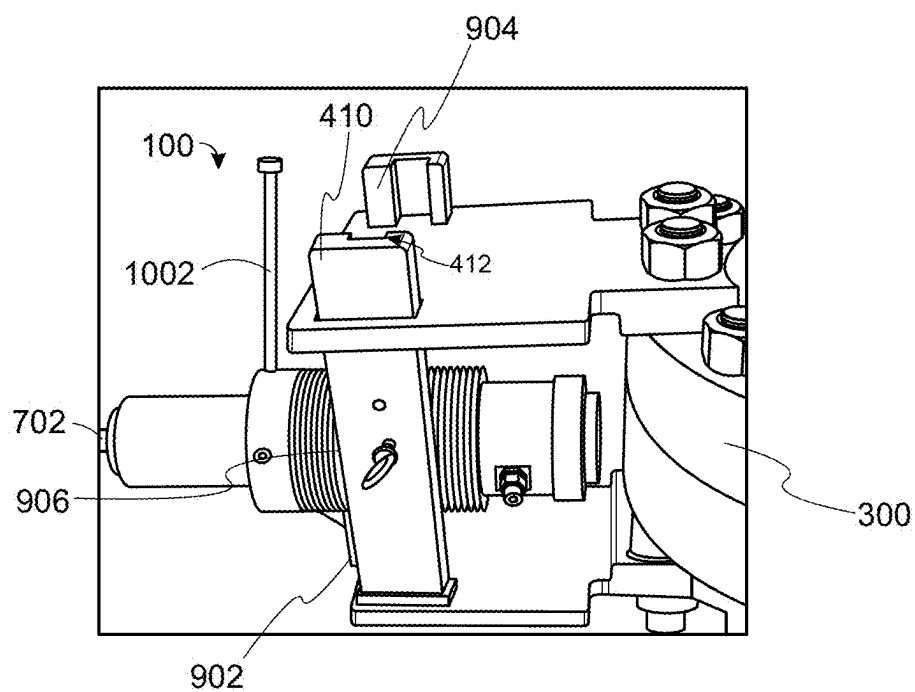


FIG. 10

CLOSED COMPARTMENT WELL TOOL FOR WELLHEAD RE-PACKING

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of U.S. Provisional Application No. 63/558,306, filed Feb. 27, 2024, the entire contents of which are incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] This disclosure pertains to operations associated with a wellhead of a production wellbore, specifically to packing/re-packing operations for oilfield wellheads.

BACKGROUND

[0003] A wellhead is a wellbore component positioned at/above a surface of the wellbore and that provides an interface to structurally position drilling and/or production components within the wellbore. The wellhead can include a variety of tools, e.g., spools, valves, adapters to both hang downhole components and to provide pressure control of the wellbore. An example wellhead, e.g., a Christmas tree, can include multiple spools, e.g., a tubing hanger spool, an intermediate casing spool, or a lower most casing spool. The intermediate and lowermost casing spools, each include a respective seal for pressure control and a slip seal to hang a casing. The tubing hanger spool can include a tubing bonnet, a repacking port, and a tubing hanger.

[0004] A wellhead spool can include an injection port (also called a pressure port) with a check valve that prevents wellbore pressure from escaping to the atmosphere through the injection port while permitting fluid flow through the port. Packing, i.e., a deformable sealing material, is inserted between the pressure port and the check valve to surround an injection screw. The packing maintains the pressure seal. A plug (also called a pipe plug or a blind plug) covers the pressure port. Operational wear and tear of the packing material can result in pressure loss through the pressure port. To avoid or overcome such pressure losses, the pressure port is re-packed, i.e., re-filled with new packing.

SUMMARY

[0005] This disclosure describes technologies relating to a closed compartment well tool for wellhead re-packing operations.

[0006] The details of one or more implementations of the subject matter described in this specification are set forth in the accompanying drawings and the description below. Other features, aspects, and advantages of the subject matter will become apparent from the description, the drawings, and the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a schematic diagram of an example of a well tool for re-packing a wellhead pressure port.

[0008] FIG. 2 is a schematic diagram of an example of a coupling subassembly of the well tool of FIG. 1.

[0009] FIG. 3 is a top view of the well tool of FIG. 1.

[0010] FIG. 4 is a lateral cross-sectional view of the well tool of FIG. 1.

[0011] FIG. 5 is an axial cross-sectional view of the well tool of FIG. 1.

[0012] FIG. 6 is a front view of the coupling subassembly of FIG. 2.

[0013] FIG. 7 is a cross-sectional view of the coupling subassembly of FIG. 6.

[0014] FIG. 8 is a schematic view of internal components of the coupling subassembly of FIG. 2.

[0015] FIG. 9 is a schematic view of the well tool of FIG. 1 coupled to an example of a wellhead.

[0016] FIG. 10 is a schematic view of the well tool of FIG. 1 partially decoupled from the wellhead.

[0017] Like reference numbers and designations in the various drawings indicate like elements.

DETAILED DESCRIPTION

[0018] During wellbore drilling and production, wellheads are routinely monitored for safety reasons and interventions are performed as needed. One focus of interventions during regular production and monitoring is to properly assess and maintain the integrity of wellhead equipment while allowing for such interventions to be contained and performed in a safe environment. The focus is to prevent exposure to dangers such as sour gas or other flammable/toxic hydrocarbon from the wellbore due to a pressure barrier loss at the wellhead.

[0019] This disclosure describes an above-surface, closed compartment well tool designed for safe wellhead pack-off repairs through available wellhead ports (also called pressure ports). Implementing the techniques described in this disclosure allows rig-less wellhead repair operations to be carried out and for remedial work to be conducted safely and efficiently without jeopardizing well integrity during wellbore production. The techniques described here can allow implementing a wellhead re-packing operation safely taking into consideration environmental and wellhead integrity issues under high pressure conditions without killing or securing the well. The tool described here can be implemented to control wellbore pressure during any operational wellhead failure or wellhead integrity problem that may result in a surface wellhead leak if not treated principally here during re-packing operations. Safety risks associated with such wellheads or with monitoring or repairing the wellhead for such wellhead leaks can be reduced by using the tool described in this disclosure.

[0020] In addition, the tool effectively isolates and contains the high-pressure of the wellbore, minimizing the risk of uncontrolled releases and blowouts. This significantly enhances the safety of personnel working on the wellhead. The tool allows operators to work at a safe distance from the wellhead, reducing their exposure to hazardous fluids and potential wellsite accidents. Additionally, the tool eliminates the need for manual intervention during pressure containment, reducing the chances of human error and associated safety risks. By effectively isolating the wellbore pressure, the tool prevents any uncontrolled release of hydrocarbons into the environment. This helps minimize the risk of spills, leaks, and contamination of surrounding soil or water resources. Overall, the use of closed compartment well tools during wellhead re-packing operations provides a range of benefits, including safeguarding personnel, protecting the environment, ensuring regulatory compliance, improving operational efficiency, reducing equipment damage, and enhancing risk management practices.

[0021] FIG. 1 is a schematic diagram of an example of a well tool 100 for re-packing a wellhead pressure port. The well tool 100 includes a coupling subassembly 102 that can couple to a pressure port positioned on an above-surface portion of a wellhead of a wellbore (not shown). The coupling subassembly 102 includes a body 104 that can couple to the pressure port on the wellhead. When coupled, the coupling subassembly 102 forms a pressure seal sufficient to prevent wellbore pressure from leaking to the atmosphere through a gap between the body 104 and the pressure port. The pressure can be a pressure of the wellbore. Alternatively, or in addition, the pressure can be pressure trapped between the back end of the pressure port and the wellhead packing assembly. The well tool 100 includes a supporting subassembly 106 that is coupled to the coupling subassembly 102. The supporting subassembly 106 includes a first plate 108 and a second plate 110. Each plate can couple to the portion of the wellhead above and below the pressure port, respectively. The supporting subassembly 106 can orient the body 104 between the first plate 108 and the second plate 110 to couple to the pressure port and to support a weight of the coupling subassembly 102.

[0022] FIG. 2 is a schematic diagram of an example of the coupling subassembly 102. The coupling subassembly 102 includes a proximal end 200 relative to the pressure port on the wellhead and a distal end 202 relative to the pressure port on the wellhead. As described below, the coupling subassembly 102 can translate towards and away from the pressure port. The proximal end 200 of the coupling subassembly 102 is in contact with and covers the pressure port when the coupling subassembly 102 forms a seal against the wellhead. Also, as described below, the surface of the proximal end 200 is shaped to match and complement the surface of the wellhead. The matching complementary shapes of the wellhead and the surface of the proximal end 200 promote the seal against the wellhead to ensure that the wellhead pressure remains trapped within the well tool 100.

[0023] FIG. 3 is a top view of the well tool 100. The well tool 100 can couple to a wellhead 300, specifically to a plug 302 that covers a pressure port formed in a circumferential surface of the wellhead 300. A distal end 202 (FIG. 2) of the coupling subassembly 102 is visible in the top view, whereas the proximal end 200 of the coupling subassembly 102 is covered by the first plate 108 of the supporting subassembly 106.

[0024] FIG. 4 is a lateral cross-sectional view of the well tool 100 along the cross-sectional line A-A. As shown in FIG. 4, the well tool 100 includes a body 402 (cylinder pusher body) formed as a cylindrical container. The body 402 is formed as a hollow cylindrical portion. Another hollow cylindrical portion 404 (inner cylinder) is positioned concentrically within the hollow cylindrical portion formed by the cylinder pusher body 402. As described below with reference to FIG. 8, the cylinder pusher body 402 can rotate independently of the inner cylinder 404. The inner cylinder 404 can translate with the cylinder pusher body 402.

[0025] In some implementations, threads can be defined on the inner surface of the cylinder pusher body 402. The threads can mate with threads defined on the outer surface of the inner cylinder 404. When the cylinder pusher body 402 moves due to rotation, the mated threads cause the inner cylinder 404 to also move with the cylinder pusher body 402. To rotate the cylinder pusher body 402, an opening 406 (cylinder pusher drive port) is formed that extends radially

from an outer surface of the cylinder pusher body 402 towards (sometimes all the way to) an inner surface of the cylinder pusher body 402. By inserting a rod into the opening 406, the cylinder pusher body 402 can be manually rotated. Also, in some implementations, threads can be defined on the outer surface of the cylinder pusher body 402. The outer threads of cylinder pusher body 402 can mate with threads defined on the inner surface of the third plate 902, as described below with reference to FIG. 9.

[0026] When the cylinder pusher body 402 is held stationary and the inner cylinder 404 is rotated, the mated threads cause the inner cylinder 404 to move relative to the cylinder pusher body 402. In some implementations, the outer surface of the inner cylinder 404 and the inner surface of the cylinder pusher body 402 can be coupled without threads. In such a construction, the inner cylinder 404 can slide (non-rotationally) relative to the cylinder pusher body 402. For example, the cylinder pusher body 402, as a whole, can be rotationally moved towards or away from the plug 302 (FIG. 3) of the wellhead 300 (FIG. 3). Further, the inner cylinder 404 can translate (i.e., be pulled or pushed) within the first hollow cylindrical portion 404 in response to an axial force applied to the inner cylinder 404. In some implementations, the inner surface of the cylinder pusher body 402 defines a shoulder 405 and the outer surface of the inner cylinder 404 defines a shoulder 407. The shoulders 405 and 407 abut against each other to limit a relative movement of the inner cylinder 404 within the hollow portion of the cylinder pusher body 402.

[0027] The supporting subassembly 106 includes a housing 408 that receives the coupling subassembly 102. Details of the housing 408 are described with reference to FIG. 9. The housing 408 can allow adjusting a position of the coupling subassembly 102 relative to the wellhead 300 (FIG. 3). For example, the housing 408 can define an opening through which the cylinder pusher body 402 can translate towards and away from the wellhead 300. The housing 408 can also include two support bars (for example, a first support bar 410) using which a location of the body 402 between the first plate 108 and the second plate 110 can be adjusted. To do so, each support bar can define a guide (for example, a first guide 412 defined by the first support bar 410) formed as a groove on the support bar, to which the coupling subassembly 102 can be mounted. The guides span between the first plate 108 and the second plate 110, and allow sliding the coupling subassembly 102 to any location on each guide between the two plates.

[0028] The supporting subassembly 106 includes two bolts 304 (for example, studs with full thread and nuts) and a support rod 414. The support rod 414 passes through the first plate 108 and the second plate 110 to which the two bolts 304 are fastened on either side of the respective plates. The support rod 414 passes through an opening formed on a flange of the wellhead 300. In this manner, the support rod 414 firmly couples the supporting subassembly 106 to the wellhead 300. The supporting subassembly 106 can include multiple such support rod-bolts arrangements (for example, at least two or more) to fasten the supporting subassembly 106 to the wellhead 300. The supporting subassembly 106 also includes shim plates 416 (for example, a shim plate for the first plate 108 and another for the second plate 110) which serves as an extender plate to accommodate the flange of the wellhead 300. For example, one edge of the shim plate 416 can have a shape that matches the shape of the corre-

sponding plate (i.e., the first plate **108** or the second plate **110**). The opposite edge of the shim plate **416** can complement the shape of the flange of the wellhead **300**.

[0029] FIG. 5 is an axial cross-sectional view of the well tool **100** along the cross-sectional line B-B. As described with reference to FIG. 4, the well tool **100** includes two support rods **414**, each of which couples the first plate **108** and the second plate **110** to the flange of the wellhead **300**. The well tool **100** includes a gland nut **502** with a ferrule sleeve that is coupled to the body **104**. The gland nut **502** allows coupling of a measuring and recording system (for example, a pressure gauge) with the well tool **100** to measure the internal pressure. To do so, the well tool **100** also includes a tube **504**, one end of which is connected to the gland nut **502** through the ferrule sleeve. An autoclave fitting **506** (for example, an autoclave tee) is connected to an opposite end of the tube **504**. The autoclave fitting **506** is a T-shaped component with three openings. One opening can be connected to a tube (described later) through which fluids (for example, grease or sealant) can be flowed into the well tool and towards the pressure port. The second opening of the autoclave fitting **506** can receive pressure gauge **508**. The third opening of the autoclave fitting **506** can be plugged (i.e., kept closed). The arrangement of the gland nut **502**, the tube **504**, the autoclave fitting **506** and the pressure gauge **508** form a fluidic connection between the wellbore and the well tool **100**. The pressure gauge **508** is rated to measure a pressure greater than the pressure within the wellbore and can be used to monitor wellbore pressure while using the well tool **100**.

[0030] FIG. 6 is a front view of the coupling subassembly **102**. FIG. 7 is a cross-sectional view of the coupling subassembly along the cross-sectional line A-A. FIG. 8 is a schematic view of internal components of the coupling subassembly **102**. The coupling subassembly **102** includes the inner cylinder **404** (FIG. 4). Within the inner cylinder **404** is a piston **702** (for example, a piston stem) that can slide within the inner cylinder **404** towards and away from a plug (for example, the plug **302** (FIG. 3)) that covers the pressure port on the wellhead (for example, the wellhead **300** (FIG. 3)). The sliding motion of the piston stem **702** allows the well tool **100** to capture the plug **302** as described below. The piston stem **702** and the inner cylinder **404** reside within the cylinder pusher body **402**. The cylinder pusher body **402**, in turn, is positioned within an opening formed in a third plate (described later) of the supporting subassembly **106**. Threads (described later) formed on an outer surface of the cylinder pusher body **402** mate with threads formed in the opening in the third plate. A rotational motion of the cylinder pusher body **402** causes the coupling subassembly **102** to translate towards and away from the wellhead **300**. An external retainer ring **720** limits a distance by which the cylinder pusher body **402** can translate.

[0031] A compression ring **704** is positioned at the proximal end **200** of the body **102**. The compression ring **704** is configured to accommodate the face of the flange of the wellhead **300**. The face of the compression ring **704** is formed to have shape that matches and complements the shape of the face of the flange of the wellhead **300**. The faces of different flanges can have different shapes. Therefore, to accommodate those different shapes, different compression rings of correspondingly different shapes can be formed. The shape of each such compression ring matches and complements the shape of a corresponding face of a wellhead

flange. Having multiple such compression rings allows choosing a compression ring based on the shape of the face of the wellhead flange.

[0032] A locking ring **706** is coupled distal of the compression ring **704** relative to the flange of the wellhead **300**. The compression ring **704** resides between the proximal end **200** of the body **102** and the locking ring **706**. The locking ring **706** and the compression ring **704** are threaded to each other, for example, via threads (not shown) or quick connect/disconnect coupling (not shown). Such a coupling allows a compression ring to be quickly and easily connected to and disconnected from the locking ring and to be switched out depending on the shape of the face of the wellhead flange. The compression ring **706** defines an O-ring groove within which resides an O-ring **712** to seal the proximal end **200** of the body **104** to the flange of the wellhead **300**, thereby covering the plug that covers the pressure port. The coupling subassembly **102** additionally includes O-rings **714**, **716** and **718**. Each O-ring serves to maintain a fluidic seal between the parts between which the respective O-rings are placed. Together, the O-rings serve to seal the wellbore pressure within the well tool **100**.

[0033] A retrieving socket head **708** is attached to a proximal end of the piston stem **702** relative to the flange of the wellhead **300**. The retrieving socket head **708** is shaped to mate with the plug to be captured and retrieved from or reattached to the wellhead. A grub screw **724** is positioned at the proximal end of the piston stem **702** relative to the flange of the wellhead **300**. The grub screw **724** drives the retrieving socket head **708** towards the plug to be captured. The piston stem **702** can be pushed within the inner cylinder **404** towards the plug. When the retrieving socket head **708** reaches and covers the plug, the piston stem **702** can be rotated (for example, counter clockwise) to disconnect the plug from the pressure port. For example, the distal end **202** of the piston stem **702** can be formed to have a shape **705** (for example, a square or hexagonal shape) to fit into an appropriate tool (for example, a wrench). By positioning the tool around the shaped, distal end **202**, an operator can rotate the piston stem **702** to disconnect the plug from the pressure port. The disconnected plug is received within the retrieving socket head **708**.

[0034] The retrieving socket head **708** with the disconnected plug can then be translated away from the wellhead, for example, by continued rotation of the tool or simply by pulling. The piston stem **702** defines a shoulder **707**. The inner cylinder **404** defines a shoulder **709**. The shoulders **707** and **709** abut against each other to limit a movement of the piston stem **702** within the inner cylinder **404**. The inner cylinder **404** defines an autoclave fitting port **711** that can receive an autoclave fitting (described above). When the piston stem **702** is pushed against the port, the autoclave fitting port **711** is closed, for example, by an outer surface of the piston stem **702** or by the retrieving socket head **708**. When the piston stem **702** with the plug is retracted such that the shoulders **707** and **709** abut, the autoclave fitting port **711** is uncovered and is open to the atmosphere. When the autoclave fitting is coupled to the autoclave fitting port **711**, any wellbore pressure released from behind the pressure port passes is sealed by the autoclave fitting. In this manner, the plug can be safely removed from the wellhead flange without pressure loss.

[0035] As described above, the autoclave fitting can be coupled to a pressure gauge to measure a wellbore pressure

after the plug has been removed. Using a tube connected to the autoclave fitting, grease can be flowed into the pressure port as part of a re-packing operation. In some instances, if a leak is detected in the pressure port, a sealant can be flowed into the pressure port. The sealant can be allowed to cure to seal the pressure port completely.

[0036] The outer surface of the piston stem **702** and the inner surface of the inner cylinder **404** define an annular space **713** that spans from the shoulder **709** to another autoclave fitting port **715** to which the autoclave fitting **506** is connected. A pressure gauge coupled to the autoclave fitting **506** can measure and monitor any wellbore pressure due to leak behind the shoulder **709**. The inner cylinder **404** also includes a locking gland nut **710** that is threaded to a distal end of the inner cylinder **404**. The locking gland nut **710** keeps the inner cylinder **404** and the piston stem **702** coupled to each other. The inner cylinder **404** defines a space in the distal region between the proximal end of the locking gland nut **710**. A stem packing set **722** is positioned in this region. The stem packing set **722** locks in wellbore pressure loss due to any leak through the annular space **713** and past the autoclave fitting port **715**. The O-ring **718** further operates to lock in the wellbore pressure.

[0037] FIG. 9 is a schematic view of the well tool **100** coupled to an example of a wellhead **300**. FIG. 10 is a schematic view of the well tool of FIG. 1 partially decoupled from the wellhead **300**. As described above, the supporting subassembly **106** includes an opening to receive the coupling subassembly **102**. To do so, the supporting subassembly **106** includes a third plate **902** that is supported by two support bars—support bar **410**, support bar **904**, each of which, as described above, defines a respective guide (for example, a groove), to which the third plate **902** is mounted. Threads **906** formed on an outer surface of the cylinder pusher body of the coupling subassembly mate with threads (not shown) on an inner surface of the opening defined by the third plate **902** such that rotation of the coupling subassembly about the threads causes a translation towards and away from the wellhead **300**.

[0038] An operator inserts a tool **1002** (for example, a rod) into the opening **406** and rotates the cylinder pusher body. The rotation causes the coupling subassembly to translate towards the wellhead **300** until the compression ring contacts the flange of the wellhead flange and forms a tight seal. By coupling a tool (not shown) to the distal end of the piston stem **702**, the operator can push the piston stem **702** within the inner cylinder towards the flange of the wellhead **300**. As described above, the piston stem **702** captures the plug in the retrieving head socket. The operator turns the piston stem **702** using the tool and removes the plug from the pressure port. The operator then pulls the piston stem **702** away from the flange causing the plug to translate away from the pressure port. The translation opens the autoclave fitting port to which the autoclave fitting is connected. The operator can monitor the pressure and/or flow fluids (e.g., wellbore fluid, grease, sealant) through the autoclave fitting coupled to the autoclave fitting port. In this manner, the operator can ensure the integrity of the wellhead, carry necessary diagnostic work, and perform wellhead re-packing operations, if needed, without exposing and compromising the pressure within the wellbore and maintaining safety. Upon completion of the re-packing operations, the operator can push the piston stem **702** towards the flange. The operator can place and tighten the plug on the pressure port. The operator can

then turn the cylinder pusher body in the opposite direction to decouple the coupling subassembly from the flange of the wellhead as shown in FIG. 10. Subsequently, the operator can remove the supporting subassembly from the flange by removing the bolts.

Implementations

[0039] Certain aspects of the subject matter described here can be implemented as a well tool. The well tool includes a coupling subassembly and a supporting subassembly coupled to the coupling subassembly. The coupling subassembly is configured to couple to a pressure port positioned on an above-surface portion of a wellhead of a wellbore. The coupling subassembly includes a body configured to sealingly couple to the pressure port on the wellhead. The supporting subassembly includes a first plate and a second plate, each configured to couple to the portion of the wellhead above and below the pressure port, respectively. The supporting subassembly is configured to orient the body between the first plate and the second plate to couple to the pressure port and to support a weight of the coupling subassembly.

[0040] An aspect combinable with any other aspect includes the following features. The body includes a first hollow cylindrical portion and a second hollow cylindrical portion concentrically positioned within the first hollow cylindrical portion. The first hollow cylindrical portion is configured to rotate independently of the second hollow cylindrical portion. The second hollow cylindrical portion is configured to translate with the first hollow cylindrical portion.

[0041] An aspect combinable with any other aspect includes the following features. An inner surface of the first hollow cylindrical portion defines a first shoulder. An outer surface of the second hollow cylindrical portion defines a second shoulder. The first shoulder and the second shoulder are configured to abut to limit a translation of the second hollow cylindrical portion within the first hollow cylindrical portion.

[0042] An aspect combinable with any other aspect includes the following features. The body includes a piston stem positioned within the second hollow cylindrical portion. The piston stem is configured to slide within the second hollow cylindrical portion towards and away from a plug that covers the pressure port.

[0043] An aspect combinable with any other aspect includes the following features. The second hollow cylindrical portion includes a locking gland nut configured to threadedly couple to the second hollow cylindrical portion and to retain the piston stem within the second hollow cylindrical portion.

[0044] An aspect combinable with any other aspect includes the following features. The body includes a retrieving socket head attached to the piston stem. The retrieving socket head is configured to mate with the plug.

[0045] An aspect combinable with any other aspect includes the following features. The body includes a locking ring attached to a proximal end of the second hollow cylindrical portion relative to the pressure port.

[0046] An aspect combinable with any other aspect includes the following features. The body includes a compression ring attached to the locking ring and configured to be positioned between the locking ring and the pressure port.

The compression ring is configured to seal to an outer surface of the portion of the wellhead at which the pressure port is positioned.

[0047] An aspect combinable with any other aspect includes the following features. The compression ring defines an O-ring groove. The compression ring further comprises an O-ring positioned in the O-ring groove. The O-ring is configured to seal the compression ring to the outer surface of the portion of the wellhead at which the pressure port is positioned.

[0048] An aspect combinable with any other aspect includes the following features. The second hollow cylindrical portion includes an autoclave fitting configured to couple to a pressure gauge configured to sense a pressure within the second hollow cylindrical portion.

[0049] An aspect combinable with any other aspect includes the following features. The first hollow cylindrical portion includes threads formed on an outer surface of the first hollow cylindrical portion. The supporting subassembly includes a third plate positioned between and perpendicular to the first plate and the second plate. The third plate defines an opening in which the threads of the first hollow cylindrical portion are received.

[0050] An aspect combinable with any other aspect includes the following features. The supporting subassembly includes a first support bar and a second support bar. The third plate is attached to the first support bar and the second support bar.

[0051] An aspect combinable with any other aspect includes the following features. Each of the first support bar and the second support bar define a respective guide to which the third plate is attached. Each respective guide is configured to allow adjusting a position of the third plate between the first plate and the second plate.

[0052] An aspect combinable with any other aspect includes the following features. The first plate defines multiple first openings through which the first plate is attached to the wellhead.

[0053] An aspect combinable with any other aspect includes the following features. The second plate defines multiple second openings through which the second plate is attached to the wellhead.

[0054] An aspect combinable with any other aspect includes the following features. The multiple first openings and the multiple second openings are configured to receive respective rods configured to be bolted to the wellhead.

[0055] Certain aspects of the subject matter described here can be implemented as a method. A first plate and a second plate of a supporting subassembly of a well tool are coupled to an above-surface portion of a wellhead of a wellbore. A coupling subassembly is coupled to the supporting subassembly and positioned between the first plate and the second plate. The coupling subassembly is coupled to a pressure port positioned on the above-surface portion of a wellhead of a wellbore to form a seal between a body of the coupling subassembly and the pressure port on the wellhead. The pressure port is positioned at a location on the portion of the wellhead between the first plate and the second plate. The seal is sufficient to prevent a pressure within the wellbore from being released to the atmosphere through the pressure port.

[0056] An aspect combinable with any other aspect includes the following features. The body of the coupling subassembly includes a first hollow cylindrical portion and

a second hollow cylindrical portion concentrically positioned within the first hollow cylindrical portion. To couple the coupling subassembly to the pressure port, after coupling the first plate and the second plate of the supporting subassembly to the portion of the wellhead, the first hollow cylindrical portion is rotated between the first plate and the second plate. In response to rotating the first hollow cylindrical portion, the second hollow cylindrical portion non-rotationally translates with the first hollow cylindrical portion towards the pressure port.

[0057] An aspect combinable with any other aspect includes the following features. To couple the coupling subassembly to the pressure port, a piston stem positioned within the second hollow cylindrical portion is slid towards a plug that covers the pressure port.

[0058] An aspect combinable with any other aspect includes the following features. To couple the coupling subassembly to the pressure port, a retrieving socket head attached to a proximal end of the piston stem relative to the plug, is positioned over the plug.

[0059] Thus, particular implementations of the subject matter have been described. Other implementations are within the scope of the following claims.

1. A well tool comprising:

a coupling subassembly configured to couple to a pressure port positioned on an above-surface portion of a wellhead of a wellbore, the coupling subassembly comprising a body configured to sealingly couple to the pressure port on the wellhead; and

a supporting subassembly coupled to the coupling subassembly, the supporting subassembly comprising a first plate and a second plate, each configured to couple to the portion of the wellhead above and below the pressure port, respectively, the supporting subassembly configured to orient the body between the first plate and the second plate to couple to the pressure port and to support a weight of the coupling subassembly.

2. The well tool of claim 1, wherein the body comprises a first hollow cylindrical portion, wherein the body comprises a second hollow cylindrical portion concentrically positioned within the first hollow cylindrical portion, the first hollow cylindrical portion configured to rotate independently of the second hollow cylindrical portion, the second hollow cylindrical portion configured to translate with the first hollow cylindrical portion.

3. The well tool of claim 2, wherein an inner surface of the first hollow cylindrical portion defines a first shoulder, wherein an outer surface of the second hollow cylindrical portion defines a second shoulder, wherein the first shoulder and the second shoulder are configured to abut to limit a translation of the second hollow cylindrical portion within the first hollow cylindrical portion.

4. The well tool of claim 2, wherein the body comprises a piston stem positioned within the second hollow cylindrical portion, the piston stem configured to slide within the second hollow cylindrical portion towards and away from a plug that covers the pressure port.

5. The well tool of claim 4, wherein the second hollow cylindrical portion comprises a locking gland nut configured to threadedly couple to the second hollow cylindrical portion and to retain the piston stem within the second hollow cylindrical portion.

6. The well tool of claim 4, wherein the body comprises a retrieving socket head attached to the piston stem, the retrieving socket head configured to mate with the plug.

7. The well tool of claim 2, wherein the body comprises a locking ring attached to a proximal end of the second hollow cylindrical portion relative to the pressure port.

8. The well tool of claim 7, wherein the body comprises a compression ring attached to the locking ring and configured to be positioned between the locking ring and the pressure port, the compression ring configured to seal to an outer surface of the portion of the wellhead at which the pressure port is positioned.

9. The well tool of claim 8, wherein the compression ring defines an O-ring groove, and wherein the compression ring further comprises an O-ring positioned in the O-ring groove, the O-ring configured to seal the compression ring to the outer surface of the portion of the wellhead at which the pressure port is positioned.

10. The well tool of claim 2, wherein the second hollow cylindrical portion comprises an autoclave fitting configured to couple to a pressure gauge configured to sense a pressure within the second hollow cylindrical portion.

11. The well tool of claim 2, wherein the first hollow cylindrical portion comprises threads formed on an outer surface of the first hollow cylindrical portion, wherein the supporting subassembly comprises a third plate positioned between and perpendicular to the first plate and the second plate, the third plate defining an opening in which the threads of the first hollow cylindrical portion are received.

12. The well tool of claim 11, wherein the supporting subassembly comprises a first support bar and a second support bar, wherein the third plate is attached to the first support bar and the second support bar.

13. The well tool of claim 12, wherein each of the first support bar and the second support bar defines a respective guide to which the third plate is attached, wherein each respective guide is configured to allow adjusting a position of the third plate between the first plate and the second plate.

14. The well tool of claim 1, wherein the first plate defines a plurality of first openings through which the first plate is attached to the wellhead.

15. The well tool of claim 14, wherein the second plate defines a plurality of second openings through which the second plate is attached to the wellhead.

16. The well tool of claim 15, wherein the plurality of first openings and the plurality of second openings are configured to receive respective rods configured to be bolted to the wellhead.

17. A method comprising:

coupling a first plate and a second plate of a supporting subassembly of a well tool to an above-surface portion of a wellhead of a wellbore, a coupling subassembly coupled to the supporting subassembly and positioned between the first plate and the second plate; and

coupling the coupling subassembly to a pressure port positioned on the above-surface portion of a wellhead of a wellbore to form a seal between a body of the coupling subassembly and the pressure port on the wellhead, the pressure port positioned at a location on the portion of the wellhead between the first plate and the second plate, the seal sufficient to prevent a pressure within the wellbore from being released to the atmosphere through the pressure port.

18. The method of claim 17, wherein the body of the coupling subassembly comprises:

a first hollow cylindrical portion; and

a second hollow cylindrical portion concentrically positioned within the first hollow cylindrical portion,

wherein coupling the coupling subassembly to the pressure port comprises, after coupling the first plate and the second plate of the supporting subassembly to the portion of the wellhead, rotating the first hollow cylindrical portion between the first plate and the second plate, wherein, in response to rotating the first hollow cylindrical portion, the second hollow cylindrical portion non-rotationally translates within the first hollow cylindrical portion towards the pressure port.

19. The method of claim 18, wherein coupling the coupling subassembly to the pressure port comprises sliding a piston stem positioned within the second hollow cylindrical portion towards a plug that covers the pressure port.

20. The method of claim 19, wherein coupling the coupling subassembly to the pressure port comprises positioning a retrieving socket head attached to a proximal end of the piston stem relative to the plug, over the plug.

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